MINING DEALER BEST PRACTICE

Piston (Pump) Clearance Measuring Tool

Maintenance and Repair	Application	Component Life Management	**Component Renewal (CRC)**	MARC Management

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1.0 Introduction

The CAT V-Tool offered by Caterpillar is unable to measure the piston end clearance on all different sizes of pistons (from Hydraulic Piston Pumps). Hence, Borusan technicians have been unable to serve a measurement to support their decision on the reusability of these pistons. This resulted in them spending unnecessary time, effort, and money to replace pistons with new ones, when the original ones may have been able to be reused. Utilizing a piston clearance measurement tool prevents these kinds of problems. The device enables technicians to check and confirm that the end play meets Caterpillar's specifications and enables them to make informed decisions. Also, technicians can easily evaluate piston clearance percentage in their own working area and decide whether they can continue to work with the piston or not. This device saves time and reduces unnecessary piston waste while increasing sustainability.

2.0 Best Practice Description

Firstly, a surface-grinded plate was designed and manufactured by the technicians. This plate's purpose is to hold the tool together and making carrying the entire device easier. The aluminum material was put on top of the plate. Aluminum was selected because it is harmless to pistons and in addition it is light, making it easy to maneuver. After that, two wedges were set up to tighten the pistons which are plugged in. O-rings are also used for tightening purpose. In the end, the comparator indicator is used for getting results.

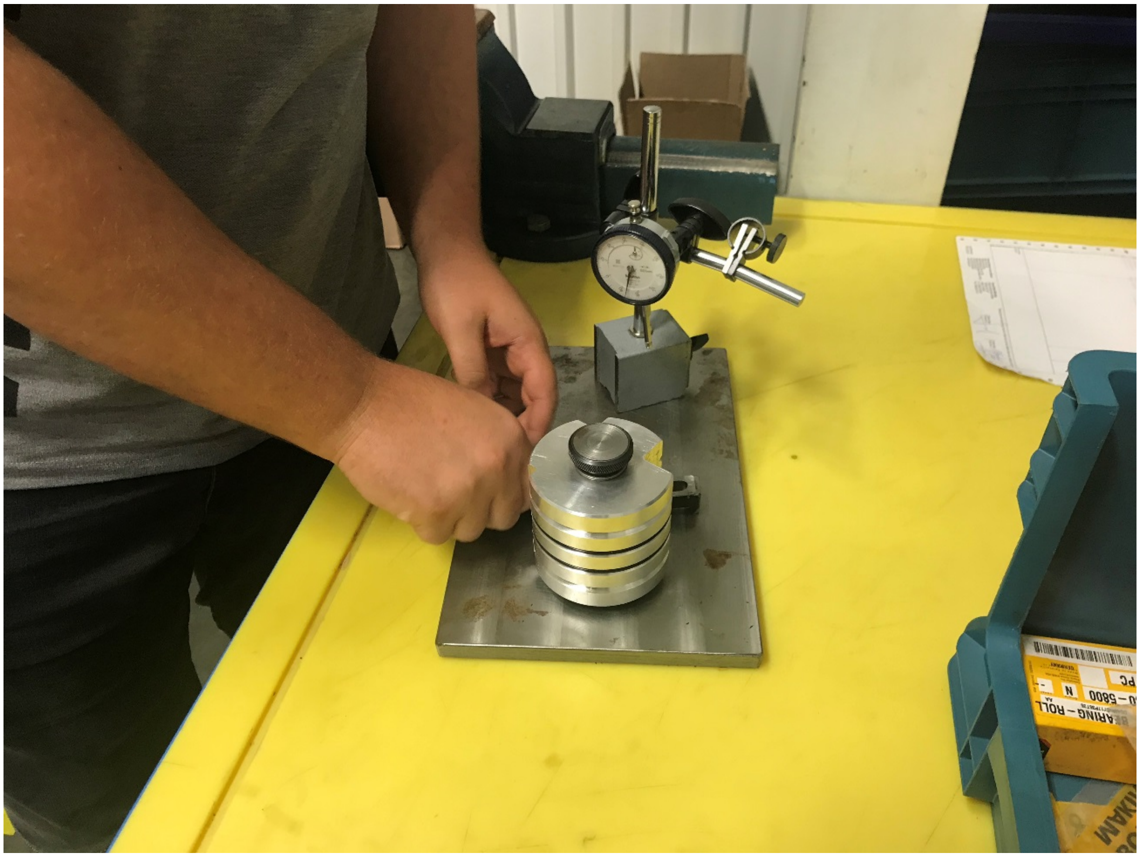
3.0 Implementation Steps

1- PLATE

A surface-grinded plate is produced. The plate should be underneath the tool.

2 - ALUMINUM MATERIAL

The aluminum material must be fixed to the plate.



3- WEDGES

The wedges should be set up into two sides of aluminum material.

4- MEASUREMENT with COMPARATOR INDICATOR

Measurements can be done with the usage of comparator indicator.

4.0 Benefits

- ✓ Technicians can easily use this tool in their own work area. This creates a time saving.
- ✓ The tool is portable and easy to use.
- ✓ Preventing unnecessary piston waste due to wrong measurements of piston clearance percentage.

5.0 Resources Required

Item	Unit Size	Cost (TL)
Aluminum Material	1	$14
Wedges	2	$7
Plate	1	$10
Comparator Indicator	1	$16
TOTAL	**5**	**$54**

6.0 Supporting Attachments / References

There are six drawings and one Application Video attached:

1. Adjustment Bolt
2. Fixture
3. Hook
4. Special Bolt
5. Table
6. Tool
7. Application Video

7.0 Related Best Practices

8.0 Acknowledgments

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